Practice Problems:
Laws and Simple Circuits

Introduction to Electric Circuits
Supplemental Instruction

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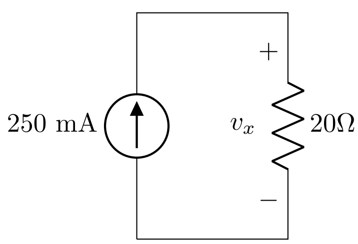
This document is not a substitute for a textbook or classroom instruction. It may contain errors and is intended to be used only to the extent that it is helpful to supplement students' learning experience.

1 Ohm's Law

Find v_x.

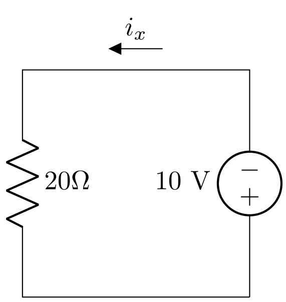


Solution

$$v_x = 250\text{mA} \cdot 20\Omega = 5 \text{ V}$$

2 Ohm's Law

Find i_x.

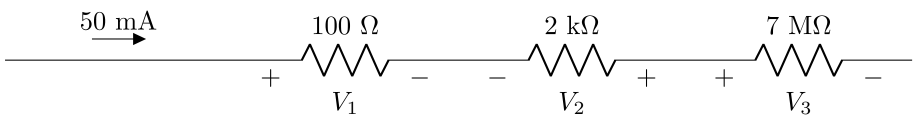


Solution

$$i_x = \frac{-10\text{V}}{20\Omega} = -500\text{mA}$$

3 Ohm's Law in Series

Find V_1, V_2, and V_3.

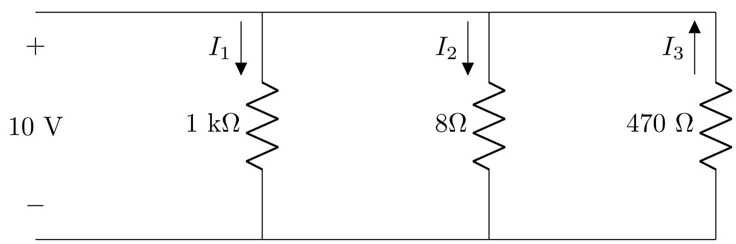


Solution

$$V_1 = 50\text{ mA} \cdot 100\ \Omega = 5\text{ V}$$
$$V_2 = -50\text{ mA} \cdot 2\text{ k}\Omega = -100\text{ V}$$
$$V_3 = 50\text{ mA} \cdot 7\text{ M}\Omega = 350\text{ kV}$$

4 Ohm's Law in Parallel

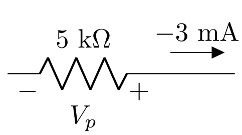
Find I_1, I_2, and I_3.



Solution

$$I_1 = \frac{10\text{ V}}{1\text{ k}\Omega} = 10\text{ mA}$$
$$I_1 = \frac{10\text{ V}}{8\ \Omega} = 1.25\text{ A}$$
$$I_1 = -\frac{10\text{ V}}{470\ \Omega} = -21.28\text{ mA}$$

5 Ohm's Law



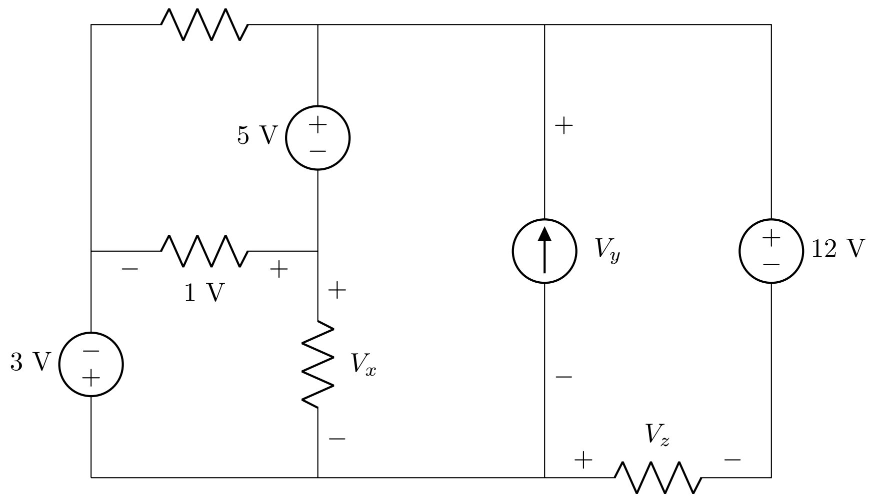
Find V_p.

Solution

$$Vp = -(-3 \text{ mA}) \cdot 5\text{k}\Omega = 15 \text{ V}$$

6 KVL

Find V_x, V_y, and V_z.

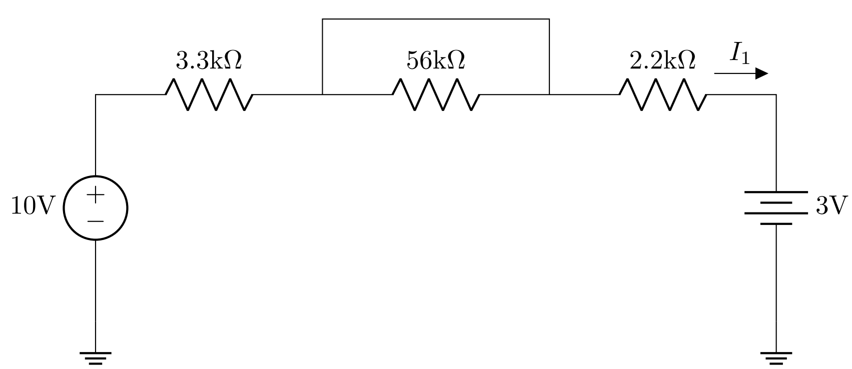


Solution

$$V_x = 1 - 3 = -2 \text{ V}$$
$$V_y = 5 + 1 - 3 = 3 \text{ V} \quad \text{or if we use } V_x, \quad V_y = 5 + (-2) = 3 \text{ V}$$
$$V_z = 3 - 1 - 5 + 12 = 9 \text{ V} \quad \text{or if we use } V_y, \quad V_z = -3 + 12 = 9 \text{ V}$$

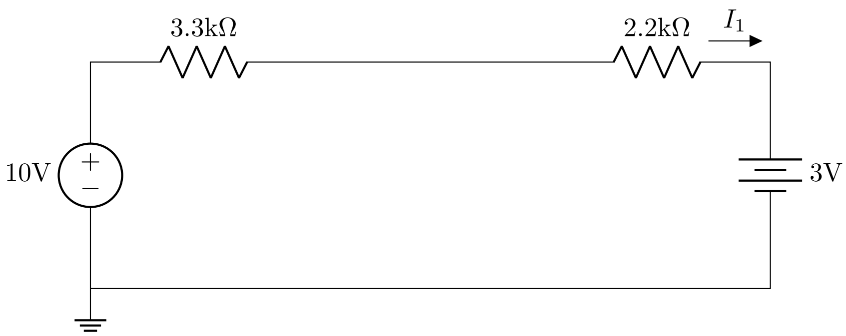
7 Finding current with KVL

Find I_1.



Solution

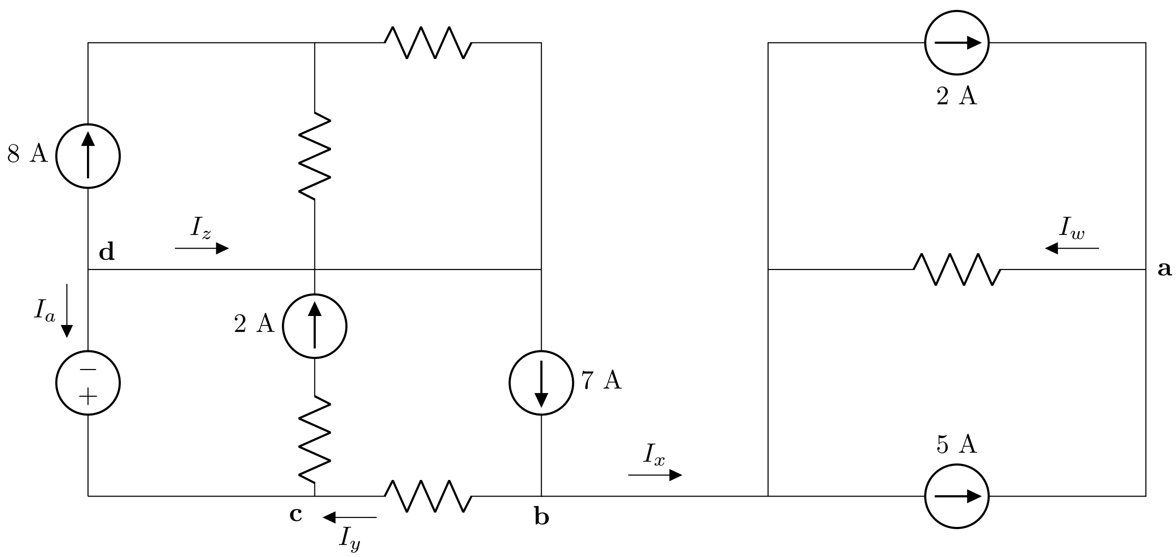
The circuit can be redrawn as:



So, $I_1 = \frac{10\text{V}-3\text{V}}{3.3\text{k}\Omega+2.2\text{k}\Omega} = 1.27$ mA

8 KCL

Find I_w, I_x, I_y, and I_z.

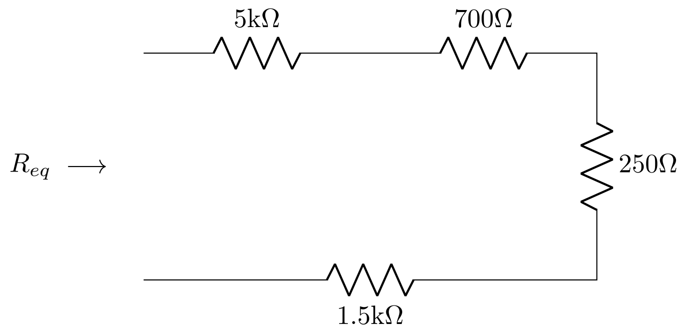


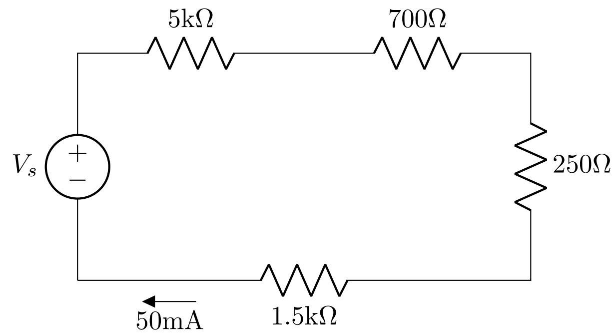
Solution

(a) Summing currents entering at **a**: $2 + 5 - I_w = 0 \Rightarrow I_w = 2 + 5 = 7$ A
(b) Summing for either half of the circuit: $I_x = 0$ A
(c) Summing currents entering at **b**: $7 - I_x - I_y = 0 \Rightarrow I_y = 7$ A
(d) Summing currents entering at **c**: $I_y + I_a - 2 = 0 \Rightarrow I_a = 2 - 7 = -5$ A
then summing currents leaving at **d**: $I_a + I_z + 8 = 0 \Rightarrow I_z = -8 - (-5) = -3$ A

9 Series Resistor Combinations

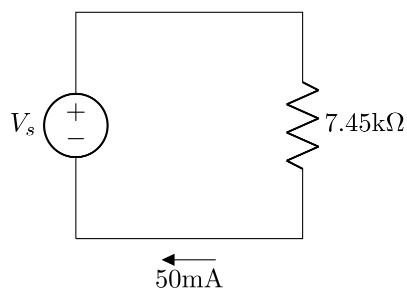
Find R_{eq} and V_s.





Solution

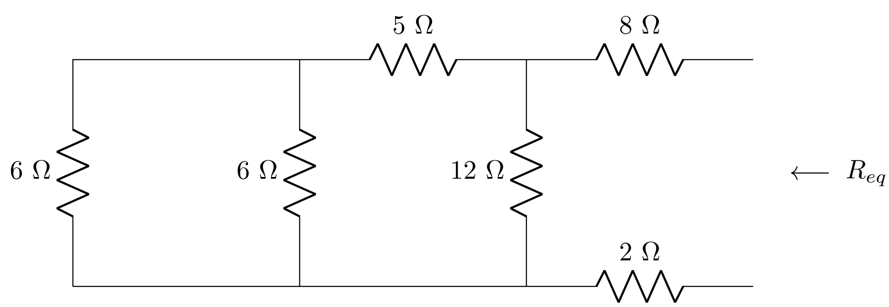
$R_{eq} = 5\text{k}\Omega + 700\Omega + 250\Omega + 1.5\text{k}\Omega = 7.45\text{k}\Omega$. We can draw the the equivalent circuit:



$V_s = 50\text{mA} \cdot 7.45\text{k}\Omega = 372.5$ V

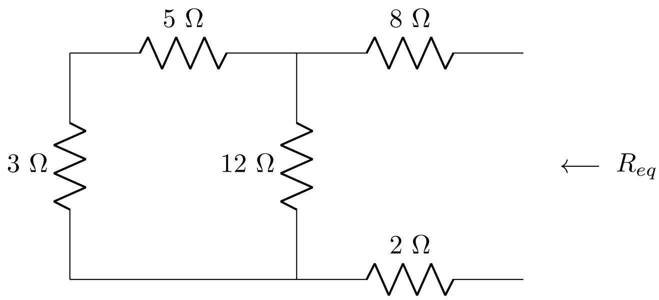
10 Resistor Combinations

Find R_{eq}.

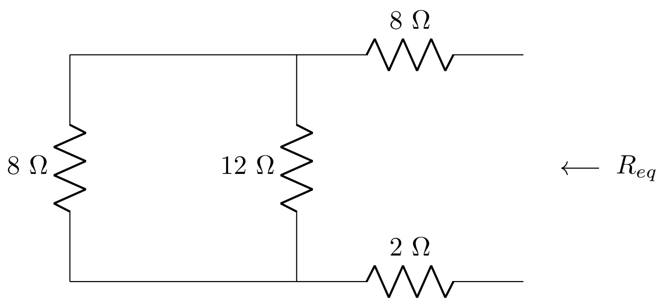


Solution

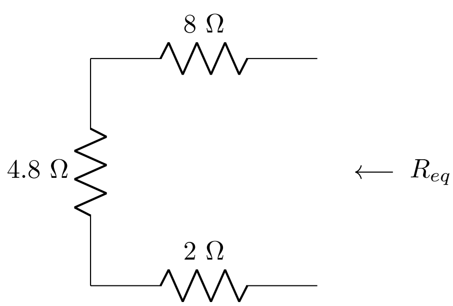
Combining the left two resistors in parallel: $\frac{1}{\frac{1}{6}+\frac{1}{6}} = 3$. We now have



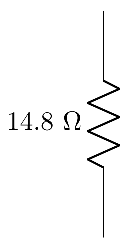
Combining the left two resistors in series, $5 + 3 = 8$. We now have



Combining the left two resistors in parallel, $\frac{1}{\frac{1}{8}+\frac{1}{12}} = 4.8$. We now have



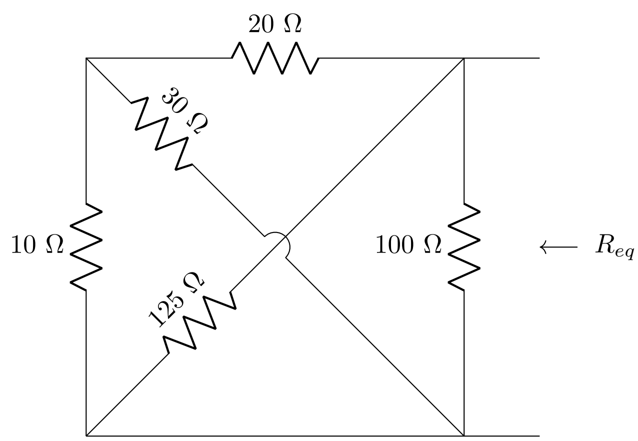
Combining the resistors in series, $8 + 4.8 + 2 = 14.8$.



$R_{eq} = 14.8\ \Omega$

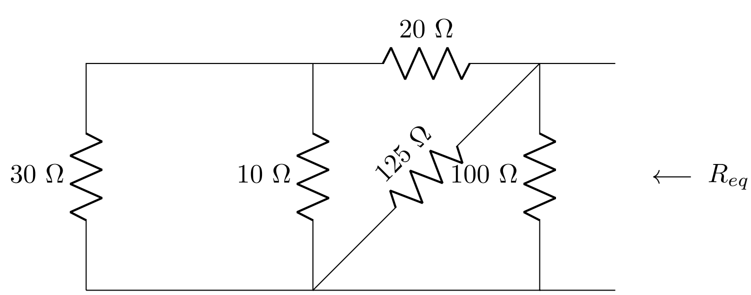
11 Resistor Combinations

Find R_{eq}.

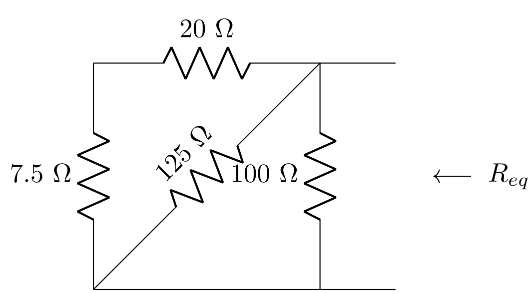


Solution

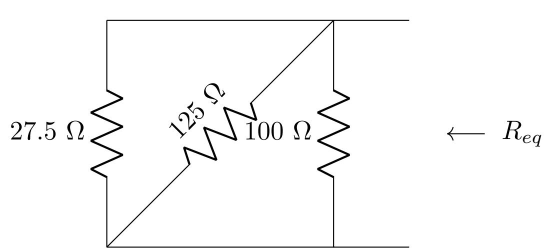
It will help to redraw the circuit by moving the 30 Ω resistor over to the left:



Combining the left two resistors in parallel, $\frac{1}{\frac{1}{30}+\frac{1}{10}} = 7.5$. We now have



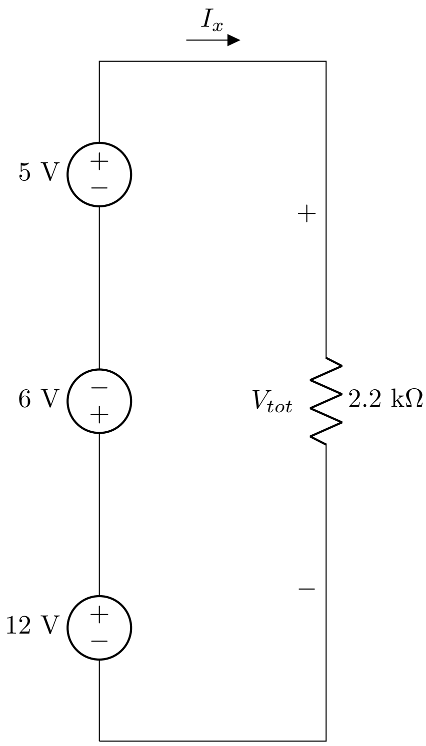
Combining the left two resistors in series, $7.5 + 20 = 27.5$. We now have



Combining the resistors in parallel, $\frac{1}{\frac{1}{27.5}+\frac{1}{125}+\frac{1}{100}} = 18.39$. $R_{eq} = 18.39\ \Omega$.

12 Series Source Combinations

Find V_{tot} and I_x.

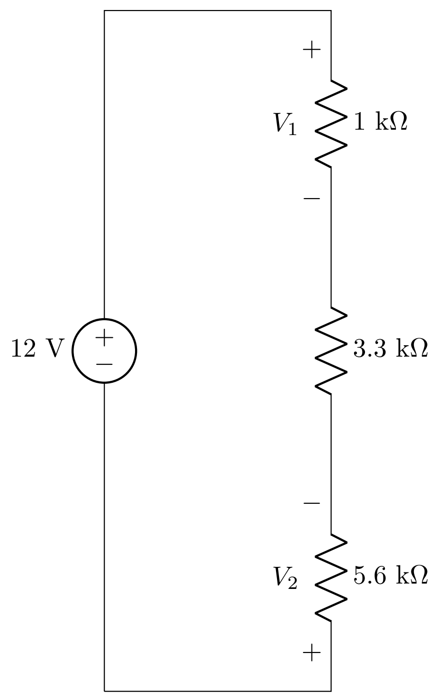


Solution

$V_{tot} = 5 - 6 + 12 = 11$ V
$I_x = \frac{V_{tot}}{2.2\text{k}\Omega} = 5$ mA

13 Voltage Division

Find V_1 and V_2.

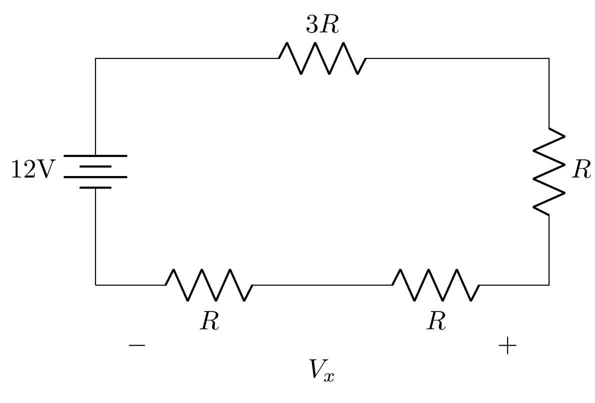


Solution

$$V_1 = 12\text{V} \cdot \frac{1\text{k}\Omega}{1\text{k}\Omega + 3.3\text{k}\Omega + 5.6\text{k}\Omega} = 1.212 \text{ V}$$
$$V_2 = -12\text{V} \cdot \frac{5.6\text{k}\Omega}{1\text{k}\Omega + 3.3\text{k}\Omega + 5.6\text{k}\Omega} = -6.79 \text{ V}$$

14 Voltage Division with unknown resistor values

Find V_x.

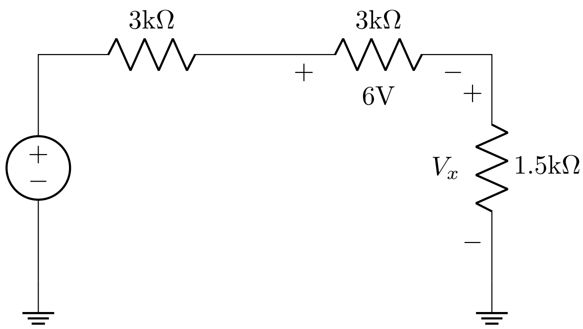


Solution

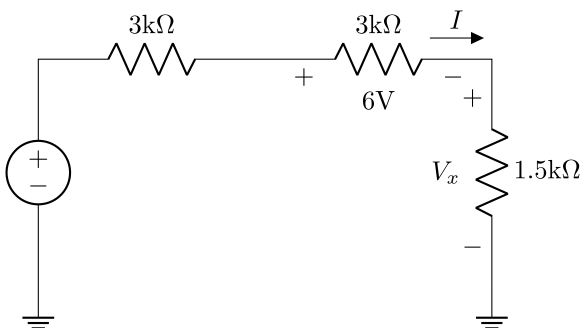
$V_x = 12\text{V} \cdot \frac{R+R}{R+R+R+3R} = 12 \cdot \frac{2}{6} = 4 \text{ V}$

15 Voltage Division / Ohm's Law

Find V_x.

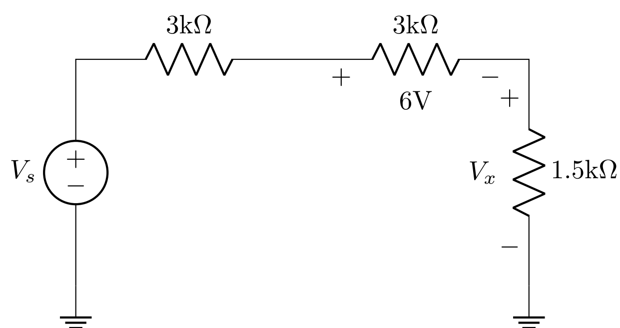


Solution (Method 1 - Ohm's Law)



Using Ohm's Law, $I = \frac{6\text{v}}{3\text{k}\Omega} = 2\text{mA}$. Then, $V_x = 2\text{mA} \cdot 1.5\text{k}\Omega = 3$ V.

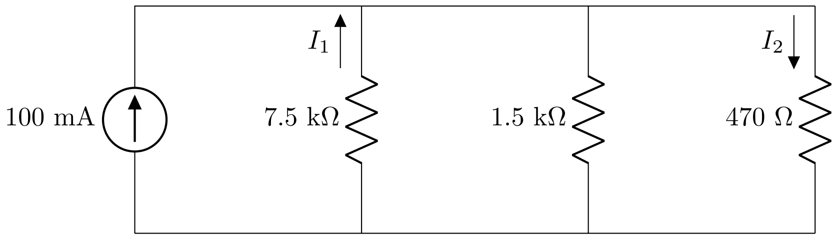
Solution (Method 2 - Voltage Division)



Using Voltage Division, $6\text{V} = V_s \cdot \frac{3\text{k}}{3\text{k}+3\text{k}+1.5\text{k}} \Rightarrow V_s = 15\text{V}$. Then, $V_x = 15\text{V} \cdot \frac{1.5\text{k}}{1.5\text{k}+3\text{k}+3\text{k}} = 3$ V.

16 Current Division

Find I_1 and I_2.



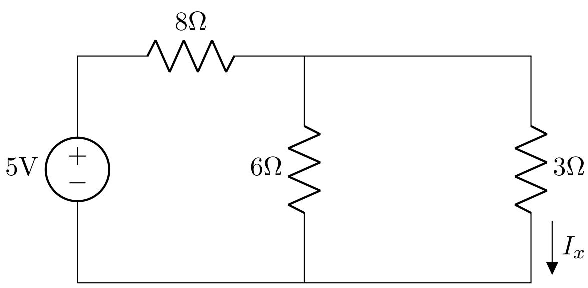
Solution

$$I_1 = -100\text{mA} \cdot \frac{\frac{1}{7.5\text{k}\Omega}}{\frac{1}{7.5\text{k}\Omega} + \frac{1}{1.5\text{k}\Omega} + \frac{1}{470\Omega}} = -4.55 \text{ mA}$$

$$I_2 = 100\text{mA} \cdot \frac{\frac{1}{470\Omega}}{\frac{1}{7.5\text{k}\Omega} + \frac{1}{1.5\text{k}\Omega} + \frac{1}{470\Omega}} = 72.67 \text{ mA}$$

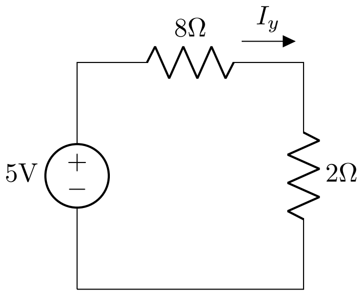
17 Voltage and Current Division with Ohm's Law

Find I_x.

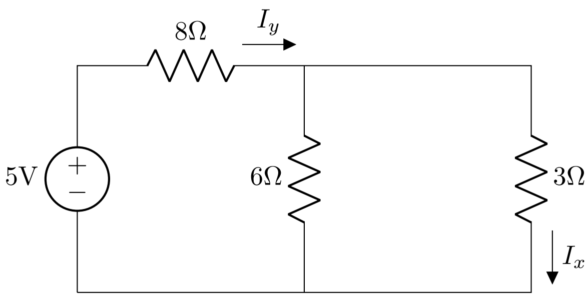


Solution (method 1)

Combining the resistors on the right: $\frac{1}{\frac{1}{6}+\frac{1}{3}} = 2\Omega$. Now we have



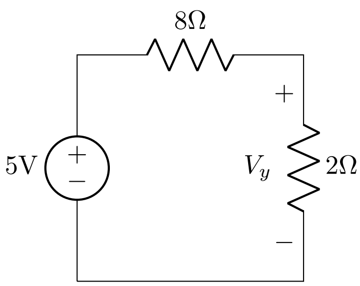
Combining resistors in series, $2+8 = 10\Omega$. Using ohm's law, we can find the total current $I_y = \frac{5\text{V}}{10\Omega} = 500\text{mA}$. We can now use current division:



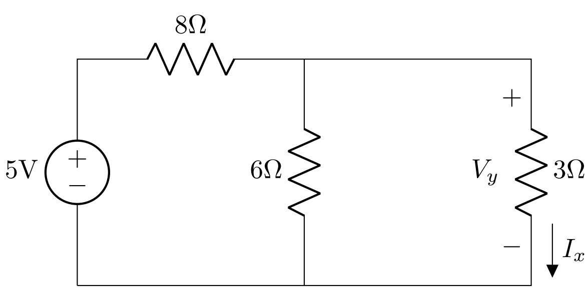
$$I_x = I_y \cdot \frac{\frac{1}{3}}{\frac{1}{3}+\frac{1}{6}} = 500\text{mA} \cdot \frac{\frac{1}{3}}{\frac{1}{3}+\frac{1}{6}} = 333.3\text{mA}$$

Solution (method 2)

Combining the resistors on the right:



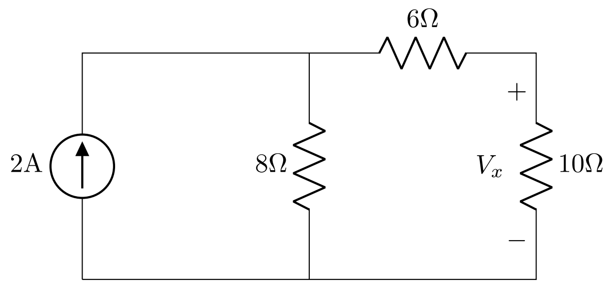
Using voltage division, $V_y = 5\text{V} \cdot \frac{2}{2+8} = 1\text{V}$. We can now use Ohm's law:



$$I_x = \frac{V_y}{3\Omega} = \frac{1\text{V}}{3\Omega} = 333.3\text{mA}$$

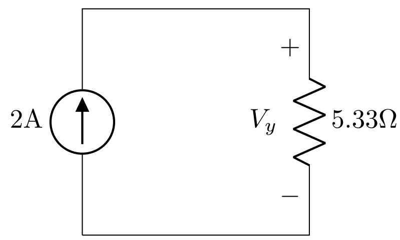
18 Voltage and Current Division with Ohm's Law

Find V_x.

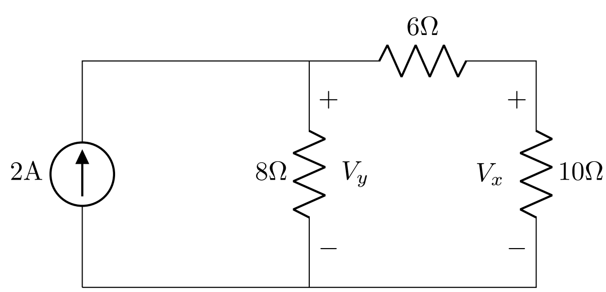


Solution

Combining resistors on the right, $6 + 10 = 16\Omega$. Then, combining resistors in parallel, $\frac{1}{\frac{1}{8}+\frac{1}{16}} = 5.33\Omega$. We now have



Using Ohm's Law, $V_y = 2\text{A} \cdot 5.33\Omega = 10.67\text{V}$. We can now use voltage division:

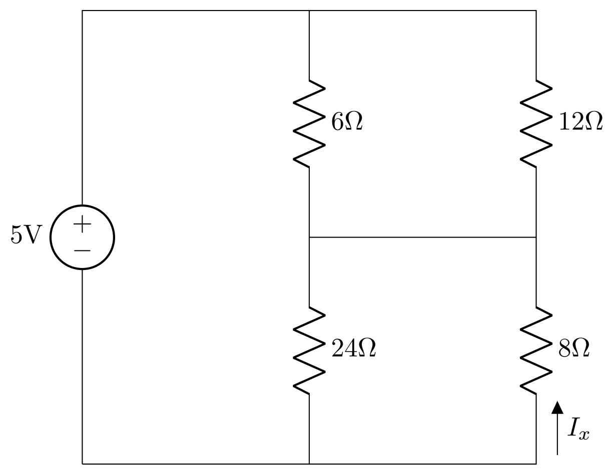


$$V_x = V_y \cdot \frac{10}{10+6} = 10.67\text{V} \cdot \frac{10}{10+6} = 6.67\text{V}$$

This problem could also be solved by combining the rightmost two resistors, using current division to find the current flowing through the rightmost branch, then using Ohm's Law to find the voltage on the 10Ω resistor.

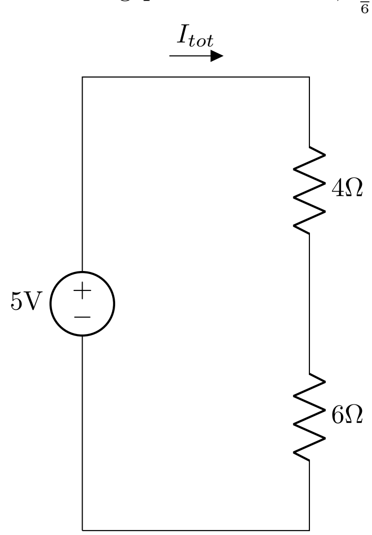
19 Current Division and Ohm's Law

Find I_x.



Solution

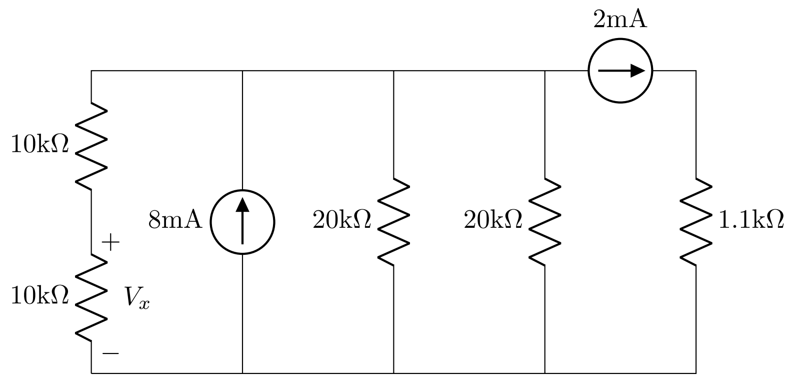
Combining parallel resistors, $\frac{1}{\frac{1}{6}+\frac{1}{12}} = 4\Omega$ (for the top), and $\frac{1}{\frac{1}{24}+\frac{1}{8}} = 6\Omega$ (for the bottom). We now have



Using Ohm's Law, $I_{tot} = \frac{5\text{V}}{(4+6)\Omega} = 500$ mA. Using current division, $I_x = -500\text{mA} \cdot \frac{24}{24+8} = -375$ mA.

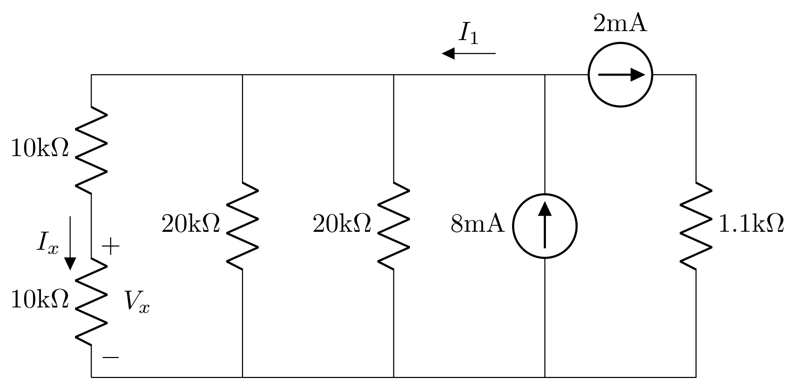
20 KCL, Current Division, Ohm's Law

Find V_x.



Solution

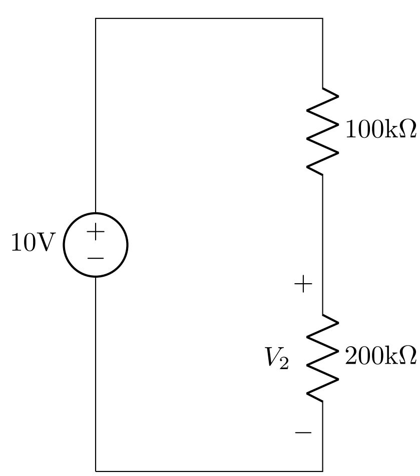
We can redraw the circuit as follows:



By KCL, $8\text{mA} = I_1 + 2\text{mA} \Rightarrow I_1 = 6\text{mA}$. Then, using current division, $I_x = \frac{6\text{mA}}{3} = 2\text{mA}$. Using Ohm's Law, $V_x = 2\text{mA} \cdot 10\text{k}\Omega = 20$ V

Loading Effect

A voltmeter with an internal resistance of 1MΩ is connected across the 200kΩ resistor. Find V_2 before and after the voltmeter is connected.

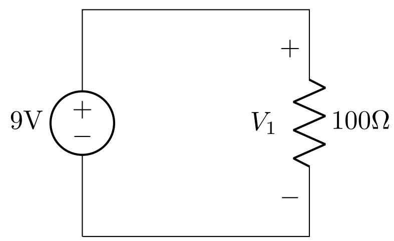


Solution

Before the voltmeter is connected, $V_2 = 10\text{V} \cdot \frac{200\text{k}\Omega}{100\text{k}\Omega + 200\text{k}\Omega} = 6.667$ V. After the voltmeter is connected, $V_2 = 10\text{V} \cdot \frac{200\text{k}\Omega || 1\text{M}\Omega}{100\text{k}\Omega + (200\text{k}\Omega || 1\text{M}\Omega)} = 6.25$ V.

21 Practical Voltage Source

Find V_1 when (a) the source has an internal resistance of 5Ω and (b) the source has an internal resistance of $10\text{m}\Omega$.

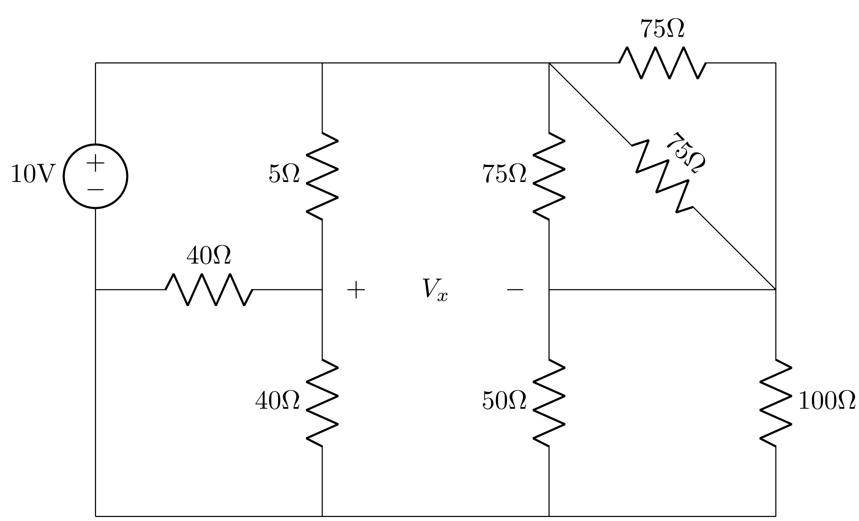


Solution

(a) $V_1 = 9\text{V} \cdot \frac{100\Omega}{100\Omega + 5\Omega} = 8.57$ V. (b) $V_1' = 9\text{V} \cdot \frac{100\Omega}{100\Omega + 10\text{m}\Omega} = 8.9991$ V.

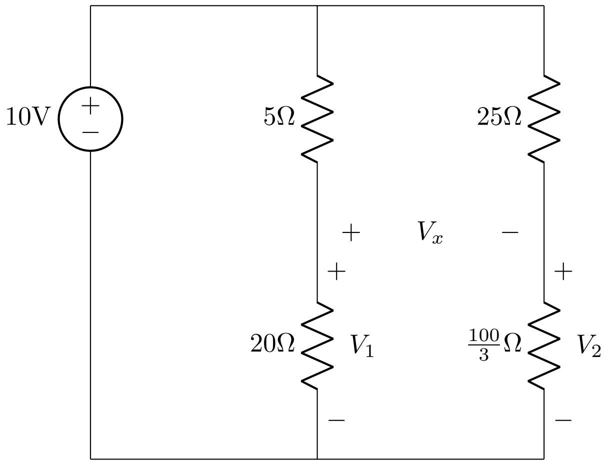
22 General Analysis

Find V_x.



Solution

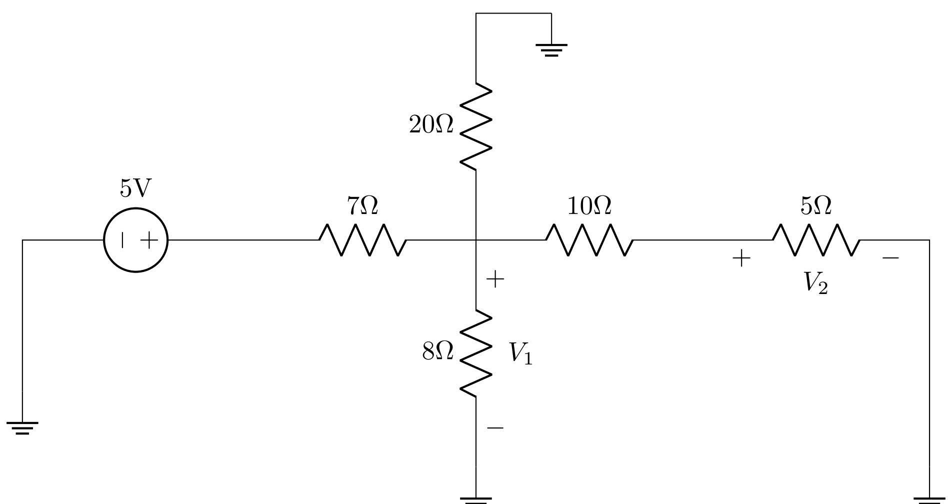
First we will combine all the parallel resistors:



We can find V_1 and V_2 using voltage division: $V_1 = 10 \cdot \frac{20}{20+5} = 8$V and $V_2 = 10 \cdot \frac{\frac{100}{3}}{\frac{100}{3}+25} = \frac{40}{7}$V. Then, using KVL, $V_x = V_1 - V_2 = 8 - \frac{40}{7} = 2.29$ V.

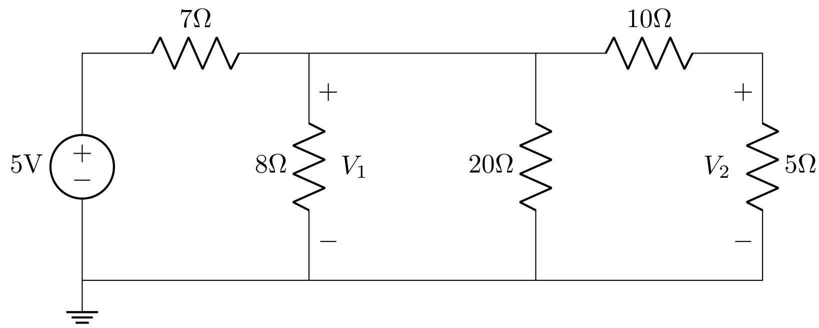
23 General Analysis

Find V_1 and V_2.

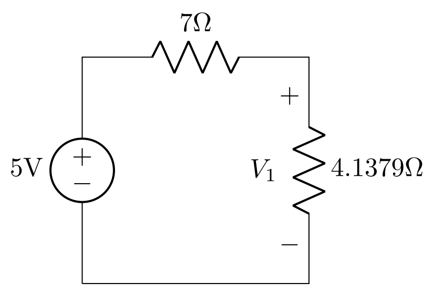


Solution

We can redraw the circuit as follows:



Combining resistors: $R_{eq} = \frac{1}{\frac{1}{8}+\frac{1}{20}+\frac{1}{10+5}} = 4.1379\ \Omega$



Using voltage division, $V_1 = 5\text{V} \cdot \frac{4.1379}{4.1379+7} = 1.858$ V. Now, applying voltage division on the original simplified circuit: $V_2 = V_1 \cdot \frac{5}{5+10} = 1.858\text{V} \cdot \frac{5}{15} = 0.62\text{V} = 620$ mV.

References

- M. Iordache. Class Lectures, Topic: “Electric circuits.” EEGR 2053, School of Engineering and Engineering Technology, LeTourneau University, 2025.
- M. Iordache. 2024. EEGR 2053 Electric circuits [PowerPoint slides]. Available: https://mviordache.name/EEGR2053/index.html
- T. L. Floyd and D. M. Buchla, Principles of Electric Circuits: Conventional Current, 10th ed. Hoboken, NJ: Pearson Education, 2020.
- O. Ortiz. 2025. ENGR 2053 Intro to electric circuits [PowerPoint slides].